

Activity No. <4.2>

<Arrays>

Course Code: CPE07

Program: Computer Engineering

Course Title: Programming logic and design

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6. Output

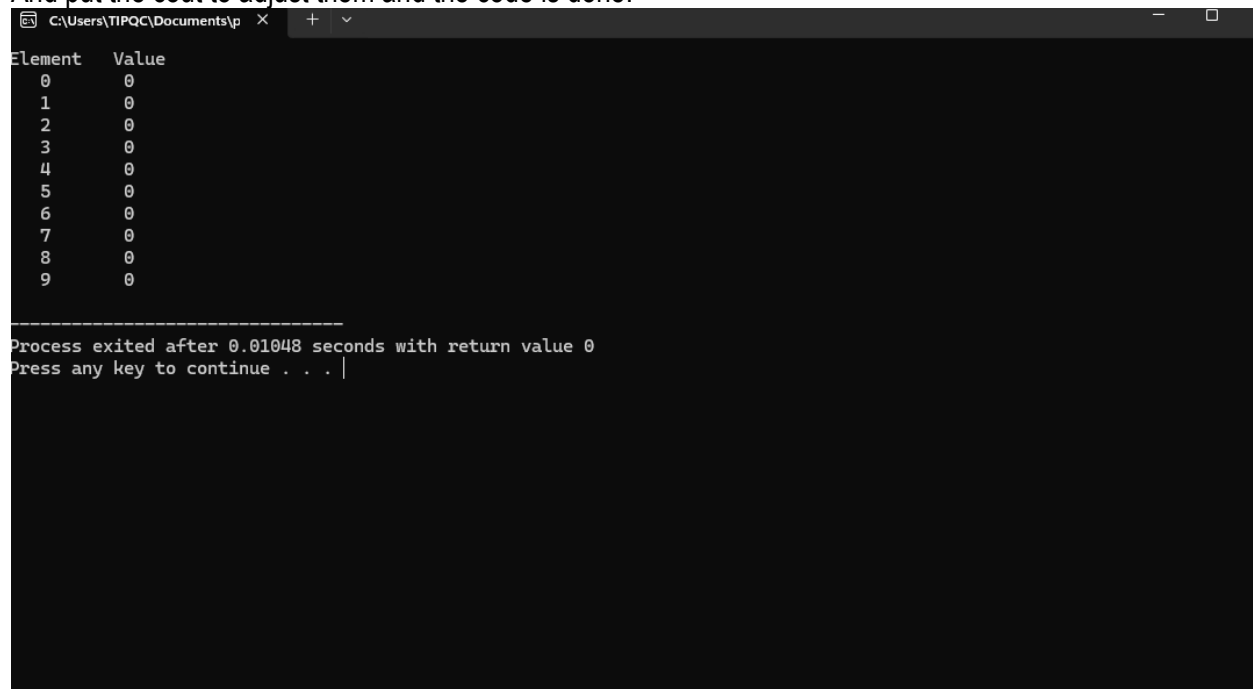
```
1  #include <iostream>
2  using namespace std;
3
4  int main () {
5      int n[10];
6
7      // Initialize array elements to 0
8      for (int i = 0; i < 10; i++) {
9          n[i] = 0;
10     }
11
12     cout << "Element   Value" << endl;
13
14     // Print index and value
15     for (int i = 0; i < 10; i++) {
16         cout << "      " << i << "      " << n[i] << endl;
17     }
18
19
20
21     return 0;
22
23 }
```

EXPLANATION:

Put your int then make the array, use the for loop to initialize the array

And put the cout to print the ELEMENT VALUE and then the other for is going to count 10 zeros

And put the cout to adjust them and the code is done.



```
C:\Users\TIPQC\Documents\p  x  +  v
Element   Value
0         0
1         0
2         0
3         0
4         0
5         0
6         0
7         0
8         0
9         0

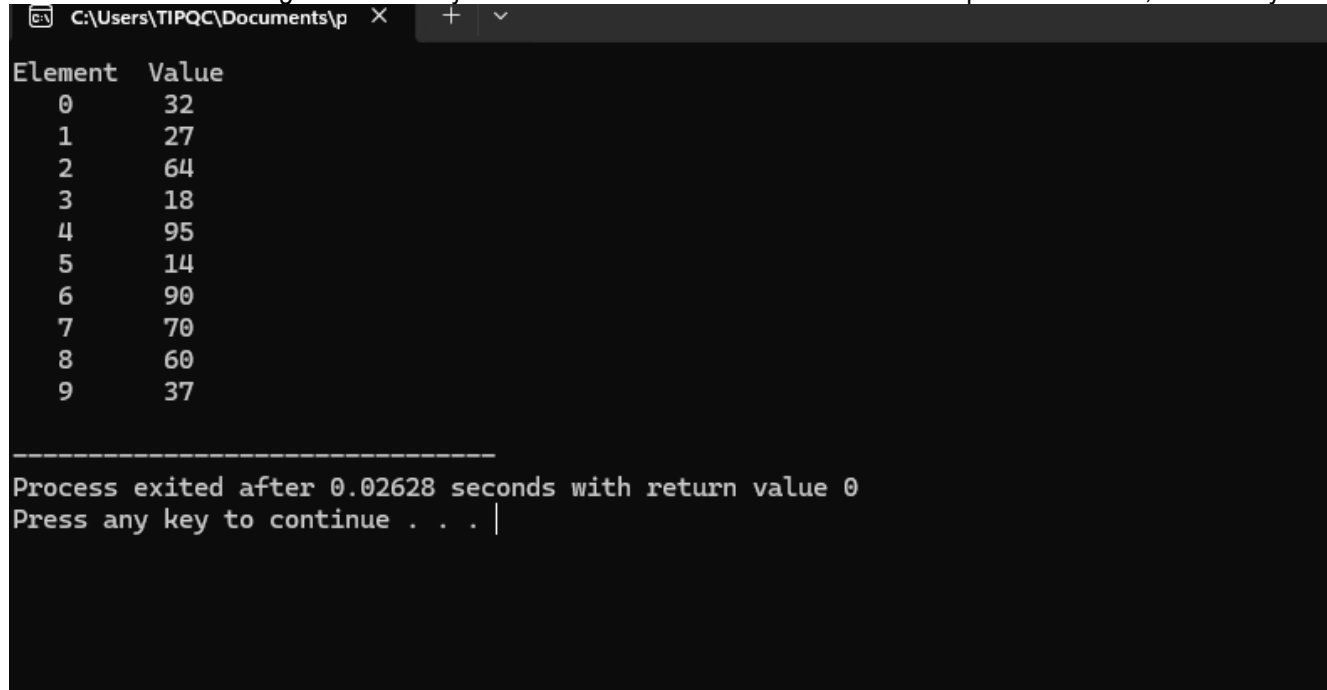
-----
Process exited after 0.01048 seconds with return value 0
Press any key to continue . . . |
```

papa.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n[10] = {32, 27, 64, 18, 95, 14, 90, 70, 60, 37};
6
7      cout << "Element Value" << endl;
8
9      for (int i = 0; i < 10; i++) {
10         cout << "    " << i << "    " << n[i] << endl;
11     }
12
13     return 0;
14 }
```

EXPLANATION:

Put an int to verify the values and put them in { } so the computer can read it and use the cout to connect the int value to make the reader don't get confused by what value name was and the other for loops counts 0 – 9, and then you're done.



```
C:\Users\TIPQC\Documents\p X + v
Element Value
0 32
1 27
2 64
3 18
4 95
5 14
6 90
7 70
8 60
9 37

-----
Process exited after 0.02628 seconds with return value 0
Press any key to continue . . . |
```

```

#include <iostream>
using namespace std;

#define SIZE 12

int main() {
    int a[SIZE] = {1, 3, 5, 4, 7, 2, 99, 16, 45, 67, 89, 45};
    int total = 0;

    for (int i = 0; i < SIZE; i++) {
        total += a[i];
    }

    cout << "Total of array element values is " << total << endl;
    return 0;
}

```

C:\Users\TIPQC\Documents\p × + ▾

```

Total of array element values is 383

-----
Process exited after 0.01797 seconds with return value 0
Press any key to continue . . . |

```

7. Supplementary Activity

```

1.
#include <iostream>
using namespace std;

int main() {
    int n [10] = {19, 3, 15, 7, 11, 9, 13, 5, 17};

    cout<< "Element\t    Value\t | Histogram" <<endl;

    for (int i = 0; i < 10; i++) {
        cout << " " << i << " " << n[i] << " ";
        for (int j = 0; j < n[i]; j++){
            cout << "*";
        }

        cout << endl;
    }

    return 0;
}

```

EXPLANATION:

First you're gonna put your int to verify the values and put the cout to have their names, Then do a for loop, then in the second "for" you will put the integer to get the values correctly and put the cout again to identify the "*" and you're done.

```
C:\Users\TIPQC\Documents\p X + v

Element    Value    Histogram
0          19    *****
1           3     ***
2          15    *****
3           7     *****
4          11    *****
5           9     *****
6          13    *****
7           5     *****
8          17    *****
9           0

-----
Process exited after 0.01468 seconds with return value 0
Press any key to continue . . . |
```

2.

```
#include <iostream>

#define responseSize 40
#define maxResponseValue 10

int main() {
    int responses[responseSize] = {1, 2, 6, 4, 8, 5, 9, 7, 8, 10, 1, 6, 3, 8, 6, 10, 3, 8, 2, 7, 6, 5, 7, 6, 8, 6, 7, 5, 6, 6, 5, 6, 7, 5, 6, 4, 8, 6, 8, 10};

    int responseCounts[maxResponseValue + 1] = {0};

    for (int i = 0; i < responseSize; i++) {
        int responseValue = responses[i];
        if (responseValue >= 1 && responseValue <= maxResponseValue) {
            responseCounts[responseValue]++;
        }
    }

    std::cout << "Response Summary:" << std::endl;
    for (int i = 1; i <= maxResponseValue; i++) {
        std::cout << "Response " << i << ": " << responseCounts[i] << " students" << std::endl;
    }

    return 0;
}
```

EXPLANATION:

First you're going to define your values and make an int to put the values inside { } and make the for loop and because there's so many numbers you can put them together in the one int,

Then put your cout to get the Response Summary, and the other for loop is gonna make the response and student separate and you'll get that output.

```
C:\Users\TIPQC\Documents\p X + v
Response Summary:
Response 1: 2 students
Response 2: 2 students
Response 3: 2 students
Response 4: 2 students
Response 5: 5 students
Response 6: 11 students
Response 7: 5 students
Response 8: 7 students
Response 9: 1 students
Response 10: 3 students

-----
Process exited after 0.02128 seconds with return value 0
Press any key to continue . . . |
```

8. Conclusion

The first one that have “*” is a little bit confusing, it took me almost 30 minutes to identify what I need to do but I realized that its just a for loop and cout at first I tried the cout only no for, and its error but when I tried it with for, it works and my problem was solved, and in the number two its kind of easy because I know the first part (for, if) But in the last part the for I was a little bit confused because I don’t put the using namespace at the first that’s why my code don’t run, but then I solved quick because its easy to find you just use the std::cout, And I learned the for loops but sometimes I’m Kinda confused on the “i” but ill work on it. And the use of the int, before I always do so many int like every value but now I learned how to make them in one int.

9. Assessment Rubric

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